

Solution Requirements

Date	29 September 2026
Team ID	SWTID-2026-5115
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define the specific functions and operations that a system must perform. In this project, they describe how ServiceNow manages users, assigns roles, and uses Access Control Lists (ACLs) to control Read, Create, Write, and Delete operations based on user roles and branch conditions.
- **Non-Functional Requirements (NFR)** define the quality and performance standards of a system. In this project, they focus on security, performance, reliability, usability, and maintainability to ensure that unauthorized access is prevented and institution records are protected.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system shall use ServiceNow user authentication to identify users requesting access to institution records.	High
2	Authorization levels	The system shall use roles bb1, bb2, bb3 and bb4 to control Read, Create, Write and Delete permissions. EEE users can access EEE branch records, while administrators have full access	High

3	External interfaces	The solution shall use the ServiceNow platform, user records, role management and the Institution Details table. No external API is required for this project.	Medium
4	Transactions processing	The system shall validate each record operation against the configured ACL rules before allowing Read, Create, Write or Delete actions.	High
5	Reporting	The administrator shall be able to inspect institution records and verify access permissions through ServiceNow lists and forms.	Medium
6	Business rules, etc.	Read access shall be restricted by the configured Branch = EEE condition and required role. Create, Write and Delete operations shall require bb2, bb3 and bb4 respectively.	High
7	Compliance to laws or regulations	The system shall restrict access to student information according to assigned permissions and the institution's applicable data-protection policies.	High
8	<i>Other</i>	The solution shall support the Institution Details table with Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number and Description fields.	Medium

Step 2: Non-Functional Requirements (NFR)

S. No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The ACL shall evaluate access requests without unnecessary delay	Normal access checks should complete within 2 seconds under test conditions.
2	Scalability	The solution should support additional institution records and authorized users within the ServiceNow instance.	Test successfully with at least 100 sample records.
3	Security & Data Privacy	The system shall prevent users from accessing records or performing operations without the required roles and conditions.	Unauthorized access is denied in all defined test cases.
4	Reliability & Availability	ACL rules shall operate consistently when users view or modify institution records.	All defined ACL test cases produce the expected result.
5	Usability & Accessibility	Administrators shall be able to configure roles and ACLs, and users shall receive the expected access or denial behavior in forms and lists	Read, Create, Write and Delete behavior is verified using test users.
6	<i>Other</i>	ACL scripts and conditions should be understandable and maintainable by ServiceNow administrators	Each ACL has a clear operation, required role and documented test result.